

Eigenvalues and inverse problems

Widom's canonical distribution conjecture for Toeplitz eigenvalues

Difficulty: extreme **Importance:** interesting to the community**Status:** Partially resolved**Topic:** Nonnormal Toeplitz eigenvalue asymptotics

Rating rationale: A general theorem must distinguish symbols whose Toeplitz spectra follow their values from the markedly different behavior of analytic symbols. This is a longstanding obstacle in the spectral analysis of nonnormal structured matrices.

Reviewed continuation — 13 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) supplies an inverse-corner criterion for one-sided annular extension and determinant limsup results. For continuous symbols with Jordan-curve range and winding $+1$ (respectively -1), absence of inner (respectively outer) extension yields a canonically distributed subsequence of the actual unperturbed Toeplitz eigenvalue measures; see the [seventh-round report](#), Theorem 2.5, Theorem 3.1 and Corollary 4.5. The Jordan-curve argument includes positive-area curves.

Status remains **Partially resolved**: subsequence convergence does not establish the displayed full-sequence conclusion, and the unrestricted continuous-symbol case remains unresolved.

The [independent informal Codex AI-agent review](#) records its accepted scope and checks. [Submission, original package, reproduction and verified affiliation](#). This is not external human peer review or formal verification; no Lean verification was performed.

Statement

Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ and $a \in C(\mathbb{T}; \mathbb{C})$. Define

$$a_k = \frac{1}{2\pi} \int_0^{2\pi} a(e^{it}) e^{-ikt} dt, \quad T_n(a) = (a_{j-k})_{j,k=0}^{n-1}.$$

Say that a extends analytically to an inner annulus if, for some $0 < r < 1$, there is a holomorphic function on $r < |z| < 1$ extending continuously to $|z| = 1$ with boundary value a . Define extension to an outer annulus analogously using $1 < |z| < R$ for some $R > 1$.

Conjecture (Widom). If a has neither such inner nor such outer extension, then for every continuous compactly supported $F : \mathbb{C} \rightarrow \mathbb{C}$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n F(\lambda_j(T_n(a))) = \frac{1}{2\pi} \int_0^{2\pi} F(a(e^{it})) dt.$$

Eigenvalues are counted with algebraic multiplicity. This conclusion is called *canonical eigenvalue distribution*. The conjecture asserts that failure requires a one-sided analytic extension; it does not assert failure for every symbol that has one.

Known cases and numerical significance

Canonical distribution holds for real-valued symbols and for the Tilli class, whose essential range has empty interior and connected complement. Widom also proved cases with nonsmooth symbols tracing Jordan curves. The general continuous-symbol assertion remains the target.

Toeplitz singular-value distributions are much better understood than their nonnormal eigenvalue counterparts. This question determines when sampling the symbol predicts the bulk eigenvalues, a basic issue for structured eigensolvers. It is distinct from [SP-06](#), which asks for real spectra in every finite order, and from individual-eigenvalue expansion problems.

References and status check

- H. Widom, *Eigenvalue distribution of nonselfadjoint Toeplitz matrices and the asymptotics of Toeplitz determinants in the case of nonvanishing index*, *Operator Theory: Advances and Applications* 48 (1990), 387–421. Historical originating source, as identified by the explicit restatements below.
- J. M. Bogoya, A. Böttcher and S. M. Grudsky, *Asymptotics of individual eigenvalues of a class of large Hessenberg Toeplitz matrices*, *OTAA* 220 (2012), 77–95. [Author manuscript](#), §1, p.2 after (1.1), states the annulus conjecture explicitly.
- M. Bogoya, S.-E. Ekström, S. Serra-Capizzano and P. Vassalos, *Matrix-less methods for the spectral approximation of large non-Hermitian Toeplitz matrices: A concise theoretical analysis and a numerical study*, *Numerical Linear Algebra with Applications* 31 (2024), e2545. [DOI](#); [institutional PDF](#). The introduction before Theorem 1 reaffirms the conjecture and states the Tilli-class result.

On 2026-09-11, checked the explicit source statements and searched Widom, canonical distribution, analytic annuli, and nonselfadjoint Toeplitz proof/counterexample combinations, including 2025–2026 results. No full resolution was found. Other conjectures bearing Widom’s name, about determinant or trace asymptotics, do not settle this eigenvalue-distribution statement. This is a bounded status check.